

AUTOMATED AUTOROTATION FOR UNMANNED ROTORCRAFT RECOVERY

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Abstract

This paper outlines the application of a real-time trajectory optimization method for guiding an autonomous unmanned rotorcraft, or a remote operator of an unmanned rotorcraft, through an autorotation in the event of total loss of power. This would allow the recovery of the unmanned rotorcraft including valuable sensor packages and data that may otherwise be lost following a power failure. Initial testing of the concept strongly indicates that successful autorotations may be performed from well within the unsafe operating area of the height-velocity profile of a helicopter by employing a fast and robust optimal algorithm that commands control motion directly in an autonomous rotorcraft or remotely through an intuitive pilot display. The algorithm generates optimal trajectories and control commands via the direct-collocation optimization method, solved using a commercially available nonlinear programming problem solver. Control inputs computed by optimal control formulation are collective and aircraft pitch, which are easily manipulated by a remote pilot or converted to collective and longitudinal cyclic commands. The formulation of the optimal control problem was carefully tailored to enable the solutions to resemble those of an expert pilot, accounting for the performance limitations of the rotorcraft as well as safety concerns.

Nomenclature

a	rotor blade two-dimensional lift curve slope (rad^{-1})	K_{ind}	induced power factor
C_P	power coefficient	m	mass of helicopter (slugs)
C_T	thrust coefficient	P_S	available shaft power ($\text{lb}\cdot\text{ft}/\text{s}$)
(C_x, C_z)	(horizontal, vertical) component of thrust coefficient	P_{res}	residual shaft power ($\text{lb}\cdot\text{ft}/\text{s}$)
c_{d_0}	mean profile drag coefficient of rotor blades	R	main rotor radius (ft)
f_e	equivalent flat plate area for fuselage (ft^2)	t_f	estimated flight time (s)
f_G	ground effect factor	(u, w)	(horizontal, vertical) velocity components (ft/s)
f_I	induced velocity factor	α	tip path plane angle (rad)
g	gravitational acceleration (ft/s^2)	γ	weighting factor in cost function
(h, d)	(vertical, horizontal) position (ft)	δ_c	collective pitch angle position (rad)
H_{hub}	rotor hub height when helicopter is on the ground (ft)	δ_{col}	normalized collective pitch angle position
I_R	polar moment of inertia of the main rotor blade ($\text{slug}\cdot\text{ft}^2$)	δ_{cyc}	normalized longitudinal cyclic position
J	cost function	η	helicopter power efficiency factor
		λ	rotor inflow ratio
		μ	rotor advance ratio
		ρ	air density (slugs/ft^3)
		σ	rotor solidity ratio
		τ_p	turboshaft engine time constant (s)
		θ	aircraft pitch angle (rad)
		Ω	main rotor angular speed (rpm)

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v	rotor induced velocity (ft/s)
v_h	induced velocity at hover (ft/s)
O_0	initial values at engine failure
$O_{\max}$	maximum value allowed
$O_{\min}$	minimum value allowed
O_{ref}	reference value
O_{nom}	reference value

Introduction

Rapid improvements in the cost/benefit ratio on pure computing processing power in the past few years have lead to the practical realization of a long-desired goal, both in Military and Civil applications, of using unmanned vehicles to perform missions that would otherwise place a human operator at risk. The advantage gained in terms of risk reduction is, however, mitigated by the reduction in the capability of an autonomous vehicle to intelligently adapt to changes in the mission environment, as would a human operator. The increased computer processing power available is, therefore, necessary to provide autonomous vehicles with a degree of intelligence that will enable them to be effective in their respective mission environments. Reaction to system failures is an example where such intelligence is required in a timely manner if the vehicle is to be recovered. A particularly challenging example is rotorcraft autorotation where quick reaction and critically timed control inputs are required for a safe landing.

Driven by cost and utility considerations, the development of unmanned aircraft is proceeding at a rapid pace. Unmanned rotorcraft are being actively developed by the USA as well as foreign militaries to accomplish missions that typically place a human pilot and crew at risk. Civil versions could also be expected as the technology develops further. Unmanned rotorcraft are also being considered for extra-terrestrial exploration. Due to cost factors, most unmanned rotorcraft under development are single-engined and would have to autorotate safely if they are to be recovered following engine failure, allowing the user to salvage not only the rotorcraft but the valuable sensor packages and data that may be on board.

Rotorcraft autorotation is a particularly challenging maneuver that requires time-critical control inputs in multiple axes to manage the energy available in the rotor and accomplish a safe landing. Typically rotorcraft manufacturers provide guidance on areas of the flight envelope (height-velocity or H-V diagram) where prolonged operation is not advised based on the capability to perform an autorotation if a total loss of

power is experienced. Rotorcraft typically transition rapidly through these “avoid” areas of the H-V diagram but some operations require prolonged flight within these areas and it may be expected that unmanned rotorcraft would be more likely to operate in these regions. A guidance methodology and cueing system that could automatically autorotate a rotorcraft or provide guidance to a remote human pilot would be beneficial in that it could potentially recover a valuable aircraft and payload that would otherwise be lost. Optimal control theory is a particularly attractive candidate for providing such guidance, as it would be able to estimate the control inputs required over a broad range of initial and final conditions associated with time-critical flight maneuvers. In an autonomous system, the optimal control inputs for autorotation computed by the optimal algorithm would simply be used to command the rotorcraft to follow the optimal trajectory. The optimal algorithm would factor in variations in vehicle weight, environmental factors, etc, and allow safe autorotation from almost anywhere in the rotorcraft’s flight envelope. Constraints would also be included in the algorithm based on performance limitations of the rotorcraft. An optimal solution could potentially exceed the capability of a skilled pilot to perform an autorotation from regions of the flight envelope that are usually avoided.

Previous Work

A series of work by Johnson (Ref. 1), Lee (Ref. 2), Zhao (Ref. 3) and Carlson (Refs. 4 and 5) applied the constrained optimization method to investigate the safe operational envelopes for autorotation and reduced-power situations for a variety of rotorcraft ranging from single-engine (OH-58A, Ref. 2) to multi-engine (UH-60A, Ref. 3) to tilt-rotor (Ref. 4). Johnson (Ref. 1) investigated the autorotation of a helicopter from a hover, and Lee (Ref. 2) refined the problem formulation by adding inequality constraints for thrust and vertical velocity. Lee postulated that the “avoid” regions in the height-velocity (H-V) restriction curve could be substantially reduced if optimal pilot inputs were used during autorotation. References 1 and 2 used a point-mass model of an OH-58A helicopter and the cost function was a weighted sum of the squared horizontal and vertical components of the helicopter velocity at touchdown. The point-mass model had two degrees-of-freedom (vertical and horizontal velocity) with an additional rotor speed degree-of-freedom. The inputs (horizontal and vertical thrust) required to minimize the cost function were computed using nonlinear optimal control theory. The correlation between flight data and the optimal results established the adequacy of the use of a point mass model in the optimal helicopter landing study (Ref. 2). Reference 2

also validated the method by comparing the optimal profiles (helicopter states and controls) with available autorotation flight-test data for the OH-58A.

A unique feature of the Ref. 2 formulation was the addition of path inequality constraints on components of both the control and the state vectors. The control variable inequality constraint is a reflection of the limited amount of thrust that is available to the pilot in the autorotation maneuver without stalling the rotor. The state variable inequality constraint is an upper bound on either the vertical sink rate of the helicopter or the rotor speed during descent. “Slack” variables were employed to convert these path inequality constraints into path equality constraints. The resultant two-point boundary-value problem with path equality constraints was successfully solved using the Sequential Gradient Restoration Algorithm (SGRA). With bounds on the control and state vectors, the optimal solutions obtained will realistically reflect the limitations of the helicopter and its pilot. The model in Ref. 2 used assumed zero-wind, vertical plane motion, and zero-slip flight.

Zhao (Ref 3) extended the work by Lee (Ref. 2) to investigate the takeoff and landing trajectories of a dual-engine helicopter in the event of a single engine failure. Zhao also used the SGRA for computing the optimal trajectories and used different constructions for the objective (cost) function to investigate optimal profiles for continued and rejected landings and takeoffs in the event of a single engine failure. In addition to touchdown velocity, horizontal distance was also included in the objective function to examine the implications of an engine failure on the safe return and landing or continued flight of the helicopter. A point-mass model of a UH-60A helicopter was used in this work with improvements to the model to include engine torque and a ground-effect model.

Carlson (Ref. 4) launched from the previous body of work and used optimal control theory to investigate the unsafe (avoid) regions of the H-V envelope in the event of single-engine failure as well as complete engine failure situations in a civil tilt-rotor aircraft. A relatively sophisticated three degree-of-freedom (vertical and horizontal velocity and pitch attitude) rotorcraft model was used with an added rotor speed degree-of-freedom and a non-linear aerodynamic model of the XV-15 tilt-rotor aircraft. An important contribution of the Ref. 4 work was the improvement in the optimization method. The Ref. 4 work demonstrated that the SGRA optimization method was not robust and efficient in the face of more complex problem formulations. Most gradient-based indirect methods, including SGRA, require longer computation time and the provision of proper initial guess values that require prior knowledge of anticipated optimal solutions. The Ref. 4 and 5 work

successfully implemented a direct method of optimization where the continuous two-point boundary value problem is discretized into a parameter optimization problem. The optimization used a well-established and mature non-linear programming problem solver that is commercially available (Ref. 6). A similar strategy was adopted in this work to compute the optimal control inputs and resulting flight path for rotorcraft autorotation.

Rotorcraft Model

The rotorcraft equations of motions are detailed below.

$$m\dot{w} = mg - \rho(\pi R^2)(\Omega R)^2 C_z - \frac{1}{2} \rho f_e w \sqrt{u^2 + w^2} \quad (1)$$

$$m\dot{u} = \rho(\pi R^2)(\Omega R)^2 C_x - \frac{1}{2} \rho f_e u \sqrt{u^2 + w^2} \quad (2)$$

$$I_R \Omega \dot{\Omega} = P_s - \frac{1}{\eta} \rho(\pi R^2)(\Omega R)^2 C_p \quad (3)$$

$$\dot{h} = -w \quad (4)$$

$$\dot{d} = u \quad (5)$$

$$\dot{P}_s = \frac{1}{\tau_p} (P_{res} - P_s) \quad (6)$$

where, P_{res} is the steady-state power remaining following a throttle cut during a simulated engine failure.

In Eq. (6) a first order response is assumed for turboshaft engines (Ref. 4). The coefficients are defined as:

$$C_p = \frac{1}{8} \sigma \epsilon_d + C_T \lambda \quad (7)$$

$$C_x = C_T \sin \alpha \quad (8)$$

$$C_z = C_T \cos \alpha \quad (9)$$

λ is the inflow ratio defined as (Ref. 4):

$$\lambda = \frac{u \sin \alpha - w \cos \alpha + v}{\Omega R} \quad (10)$$

and the induced velocity v is approximated as:

$$v = K_{ind} v_h f_l f_G \quad (11)$$

v_h is the reference induced velocity at hover defined as:

$$v_h = (\Omega R) \sqrt{\left(\frac{C_T}{2}\right)} \quad (12)$$

The induced velocity parameter f_i is defined as the ratio of the actual induced velocity to the reference velocity v_h . The following expression is used to determine f_i :

$$f_i = \begin{cases} 1/\sqrt{(b^2 + (a + f_i)^2)} & \text{If } (2a + 3)^2 + b^2 \geq 1.0; \\ a(.373a^2 + .598b^2 - 1.991) & \text{otherwise} \end{cases} \quad (13)$$

where, a and b are defined as:

$$a = \frac{u \sin \alpha - w \cos \alpha}{v_h} \quad (14)$$

$$b = \frac{u \cos \alpha + w \sin \alpha}{v_h} \quad (15)$$

The term f_G accounts for the decrease in induced velocity due to ground effect. The source model (Ref. 4) appears as:

$$f_G = 1 - \frac{R^2 \cos^2 \theta_w}{16(h + H_R)^2} \quad (16)$$

where,

$$\cos^2 \theta_w = \frac{(-wC_T + vC_z)^2}{(-wC_T + vC_z)^2 + (uC_T + vC_x)^2} \quad (17)$$

The tip path plane angle α and the aircraft pitch angle θ are effectively equivalent for the purposes of aircraft control. The collective pitch, computed using blade element theory (Ref. 2), appears as

$$\delta_c = \frac{(1 + \frac{3}{2}\mu^2)(\frac{6C_T}{a\sigma}) + \frac{3}{2}\lambda(1 - \frac{1}{2}\mu^2)}{(1 - \mu^2 + \frac{9}{4}\mu^4)} \quad (18)$$

where σ and a are the rotor solidity ratio and rotor blade two dimensional lift curve slope respectively. The advance ratio μ is defined as

$$\mu = \frac{u \cos \alpha + w \sin \alpha}{\Omega R} \quad (19)$$

Optimal Autorotation Problem Formulation

A direct method of optimization was used following the work done by Carlson in Ref. 4. In the direct method the two-point boundary value problem is transformed into a parameter optimization problem. In this formulation the states and controls are the parameters to be solved satisfying the dynamics and other physical limitations at discrete points in time (nodes), which can be solved using standard non-linear programming methods and software. The direct collocation method is used where both the rotorcraft states and controls are discretized throughout time and the rotorcraft equations-of-motion are imposed as a set of non-linear equality constraints at each point in time (or node). Based on the experience documented in Ref. 4, this method has a better convergence radius with a wider range of initial guesses (more robust to initial guess values) than other parameterization methods. The disadvantage of this method is that the dimension of the problem becomes large due to the discretization of the states and control at each node. As in Ref. 5, the parameter optimization problem was solved using the Sequential Quadratic Programming (SQP) algorithm as implemented in the SNOPT software package (Ref. 6).

Constraints

(a) Equality Constraints

1. Initial Value Constraints

States: $(\theta, u_0, \Omega_0, h_0, \dot{\theta}, P_0)$

Controls: (C_{T_0}, α_0)

2. Final Value Constraints

States: $(\infty, \infty, \infty, \theta, \infty, \infty)$

3. Equations of Motion at each node

(b) Inequality Constraints

1. State Constraints

$$-w_{\max} \leq w \leq w_{\max}$$

$$0 \leq u \leq \infty$$

$$\Omega_{\min} \leq \Omega \leq \Omega_{\max}$$

$$0 \leq h \leq \infty$$

$$0 \leq d \leq \infty$$

$$0 \leq P_s \leq \infty$$

2. Controls Constraints:

$$C_{T_{\min}} \leq C_T \leq C_{T_{\max}}$$

$$\alpha_{\min} \leq \alpha \leq \alpha_{\max}$$

The above constraints on states and controls are defined by

$$w_{\max} = 60 \text{ fps}$$

$$\Omega_{\min} = 0.75\Omega_0$$

$$\Omega_{\max} = 1.05\Omega_0$$

$$\alpha_{\min} = -20 \text{ deg}$$

$$\alpha_{\max} = 34 \text{ deg}$$

where, $\alpha_{\min}$ and $\alpha_{\max}$ are chosen as the minimum and maximum pitch values observed in flight test data (Refs.

7 and 8). $C_{T_{\min}}$ and $C_{T_{\max}}$ are aircraft-specific, with $C_{T_{\min}}$ associated with the minimum collective pitch, and $C_{T_{\max}}$ associated with blade stall. Also, to impose realistic collective range, the collective bounds are implemented such that:

$$\delta_{col_{\min}} \leq \delta_{col} \leq \delta_{col_{\max}}$$

The conversion between δ_{col} and C_T has been performed via Eq. (18) and an iterative method based on trim estimation. The constraint on the pitch angle near the ground has been imposed to prevent the tail from hitting the ground. The constraint is the function of aircraft geometry, such as the tailboom length, and altitude, and, as a result, the optimal solution guarantees that the aircraft's tail doesn't hit the ground at the final touchdown.

Objective Function:

The objective function is the sum of weighted penalties consisting of forward speed and sink rate at the final touchdown as well as the control rates for thrust coefficients and tip path angles at each node. The minimization of control rates allows smother and consistent behavior of optimal solutions.

$$\begin{aligned} J = & Q_1 \sum_{i=1}^{N-1} \left[\frac{C_T(i+1) - C_T(i)}{\Delta t} \right]^2 \\ & + Q_2 \sum_{i=1}^{N-1} \left[\frac{\alpha(i+1) - \alpha(i)}{\Delta t} \right]^2 \\ & + Q_3 (u_{t_f})^2 + Q_3 (w_{t_f})^2 \end{aligned} \quad (20)$$

where, i is the node number (where $i = 1$ is the first node at $t = 0$) and Q_i represents proper weighting factors that is selective for best performance.

Validation of the algorithm using flight data was presented previously in Ref. 9 and showed that the optimal trajectories computed with this formulation were reasonable when compared with those accomplished by an expert pilot in flight tests.

Flare Law:

In real-time application for automated autorotation, performance differences between the rotorcraft dynamics and the point-mass model used in the optimization as well as simulation timing issues cause a mismatch in the altitude predicted by the optimization algorithm and the actual altitude of the rotorcraft (simulation, in this case). During initial development it was noticed that this mismatch caused the rotorcraft to flare too early or too late. To compensate for these deficiencies, a flare law was devised that would take over from the optimal guidance at a pre-determined altitude near the ground and flare the rotorcraft based on a more conventional compensatory control law. In practical terms, this flare law attempted to recreate the final flare and landing performed by a pilot based on outside visual cues. The purpose of the optimal trajectory was to bring the rotorcraft to a pre-flare altitude at an energy condition that was conducive to a safe flare and landing.

The flare law was activated at a height of approximately 30 ft above ground and used a non-linear algorithm to modulate airspeed through pitch attitude and rotor-speed and sink-rate through collective control. The activation altitude required some adjustment during development and evaluation to compensate for the variations in aircraft weight.

Simulator Description:

Development and evaluation of the automatic autorotation and autorotation flight director display took place on a commercial helicopter Flight Training Device (FTD) manufactured by Frasca International, Urbana, IL. Although not officially certified, the FTD used for the evaluation incorporated a level of fidelity necessary for achieving FAA Certification as a Level 4 FTD. The FTD was a fixed-base simulation of a Bell-206L-4 single-turbine, single rotor helicopter (Fig. 1) with a realistic reproduction of the cockpit with a frame and dual controls and a dome visual system with 180-deg horizontal and 60-deg vertical visual field-of-view (Fig. 2). An additional graphics channel provided visual imagery immediately below the cockpit door and through the chin window on the pilot's side. The

cockpit controllers were replicas of the actual cyclic, collective and pedal controls and had realistic feel.



Figure 1. Bell 206L-4 single rotor helicopter



Figure 2. Frasca International Bell 206 Flight Training Device (FTD)

Complete engine failures could be triggered from the simulator operator's station at any time. Engine failures resulted in immediate loss of all engine power and the activation of appropriate warning lights and audio alarms. A low-rotor RPM warning light was also provided. The rotorcraft simulation model was a rotor disk model with aerodynamic models for the fuselage and empennage surfaces. The rotorcraft model had previously been evaluated by line pilots as part of the FTD acceptance testing and found to be representative

of the actual aircraft in the regular and autorotation flight regimes. The primary development pilot for this project, Ed Bachelder, an experienced helicopter pilot (SH-60B pilot) also found the rotorcraft simulation to be realistic.

Implementation of Optimal Guidance Algorithm

The block diagram in Figure 3 indicates how a Laptop PC with the real-time optimization algorithm was linked with the Frasca simulation computer.

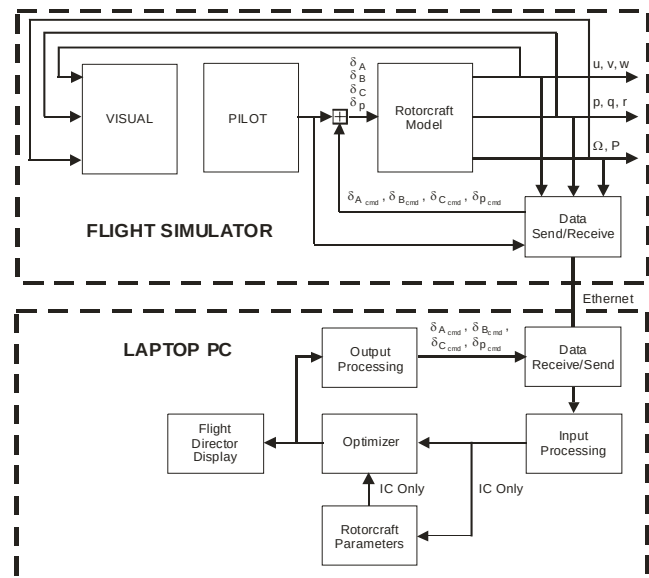


Figure 3. Interface between the optimal guidance and the FTD

The PC used for the development and evaluation of the optimal guidance was a standard commercial laptop PC with a 2 GHz Intel Pentium® processor and a Windows 2000® operating system. The PC accepted rotorcraft state and control information at a nominal 30Hz data rate and output collective, cyclic, and pedal control positions to the simulation computer, also at a 30Hz data rate. Communication was facilitated through an Ethernet link using standard Microsoft Windows compatible communication protocol. During powered flight, the optimal algorithm continuously updated the optimal solution based on the rotorcraft states (primarily speed and altitude) being received from the simulation computer. In effect, the optimizer continuously computed an updated optimal trajectory for autorotation with the assumption that an engine failure had just occurred. Typically, a new update was available every 3 sec or sooner. As implemented, when an engine failure occurred, the automatic autorotation guidance was based on the last optimal trajectory update that was available. In the future, the optimal trajectory will be updated throughout the autorotation

maneuver. The optimal algorithm considered only the optimal trajectory in the longitudinal axis (collective and longitudinal cyclic commands only). For the development and evaluation process, simple compensatory feedback control was implemented to maintain roll attitude and heading via lateral cyclic and pedal commands.

A guidance display was also generated on the laptop computer that provided an indication of how well the helicopter was following the optimal guidance during automatic autorotations. For piloted operations, such as with a remote operator, this display could be used as a flight director to guide the operator on the optimal control inputs required to accomplish a safe landing. The guidance display included a novel display concept that guided a human operator on following the optimal control inputs by providing a preview of the complete trajectory. The details of this display concept are not included in this paper due to its primary focus on autonomous operation.

The primary intent of the development and validation of the optimal guidance algorithm in this real-time simulation environment was to evaluate the robustness of the guidance algorithm across the flight envelop of the simulated helicopter. Invariably, however, emphasis was placed on the “worst case” flight conditions; i.e., entry into autorotation from flight conditions that are well within the “avoid” region of the height-velocity diagram for this helicopter (shown in Fig. 4) as these clearly illustrate the benefit of the optimal guidance. Development and refinement of the algorithm and its real-time mechanization at these flight conditions would also maximize the probability that the guidance would enable safe autorotations from flight conditions outside the avoid region. The majority of the development and evaluation was performed at a light-weight condition with limited evaluations at the heavy (maximum gross weight) and medium weight conditions.

The optimal control algorithm used a simple point-mass type model for the rotorcraft. For the algorithm to provide appropriate autorotation guidance, therefore, it was necessary to fine-tune the point-mass model parameters such that the dynamics and performance of the point-mass model approximated the rotorcraft model as implemented in the simulator as closely as possible. For automated autorotations, it was particularly important to scale and bias the optimal control inputs computed by the optimal algorithm so that it would be able to backdrive the simulation correctly. An automated procedure was setup using Matlab® to facilitate this parameter optimization process using rotorcraft state and control time history data obtained from the simulator.

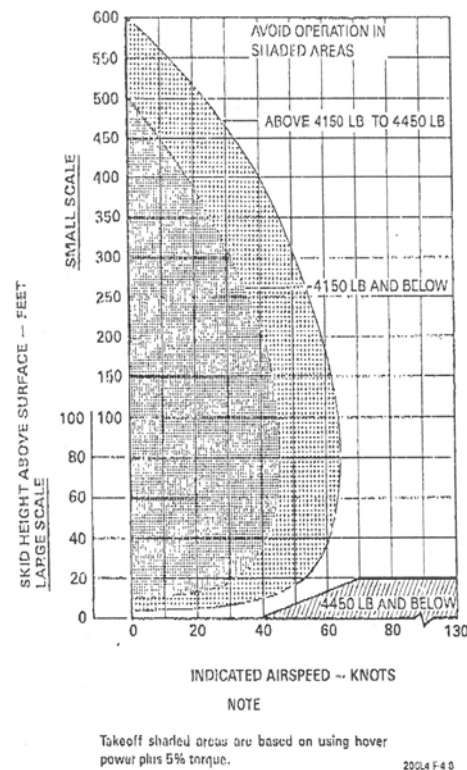


Figure 4. Height-Velocity diagram for the Bell 206L-4 Helicopter

Results

Following three-week period of development on the Frasca FTD in Urbana, IL, the automatic autorotation capability was refined to an extent that allowed evaluation of the algorithm over a range of autorotation entry conditions. The entry conditions that were attempted at light (2900 lbs), medium (3500 lbs), and heavy (4450 lbs) weight configurations using the automatic autorotation guidance are presented on a height-velocity diagram in Fig. 5. The manufacturer’s height-velocity “avoid” regions are indicated on the figure. Successful landings are shown as clear symbols and crash landings are shown as solid symbols. Crash landings represent those where the touchdown sink-rate or forward speed exceeded the limitations for the rotorcraft. Tail-strikes were also counted as crash landings. The determination of a safe or crash landing was made by the Frasca simulation software and crash landings were visually indicated by a sudden rotation and loss of the visual display.

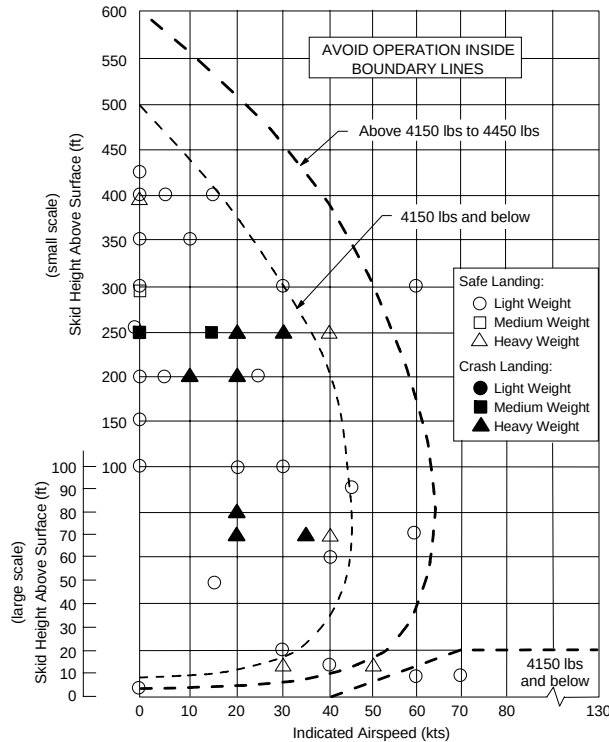


Figure 5. Automated autorotation flight conditions evaluated

As may be observed from Fig. 5, it was clearly established that using optimal guidance, safe autorotations were possible from well inside the “avoid” regions of the H-V curve including the high-speed region. Fewer evaluations were conducted at the medium and heavy weight conditions which allowed little time for adjusting the optimal algorithm parameters to ascertain whether crash landings at these weight conditions were due to performance limitations of the rotorcraft or the non-optimal formulation of the optimization problem. For example, at the heavy and medium weight conditions, refining the constraints (rotor-speed and vertical speed limits, for example) as well as the flare law parameters may have allowed greater success than was achieved. Nevertheless, safe landings were accomplished at these weight conditions from well within the “avoid” regions of the H-V curve for these weights, although not with the consistency that was achieved at the light-weight condition.

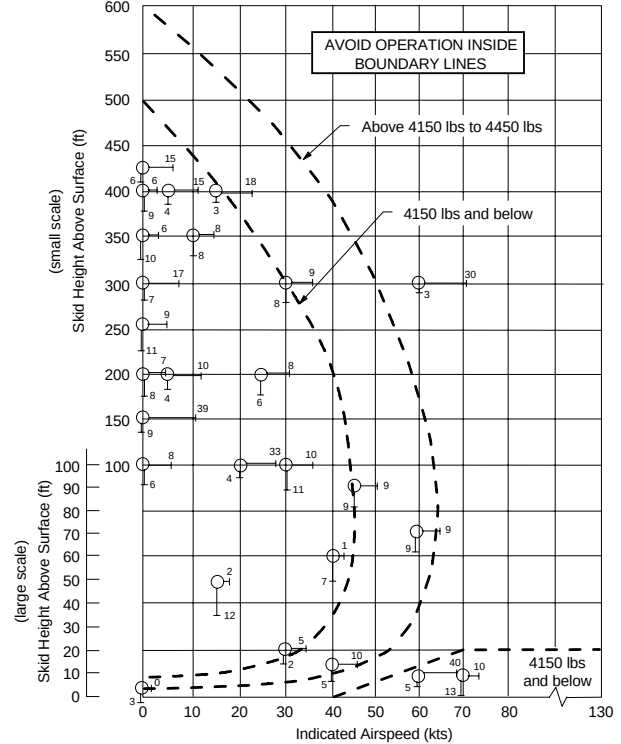


Figure 6. Touchdown ground-speed and sink-rate (light weight condition)

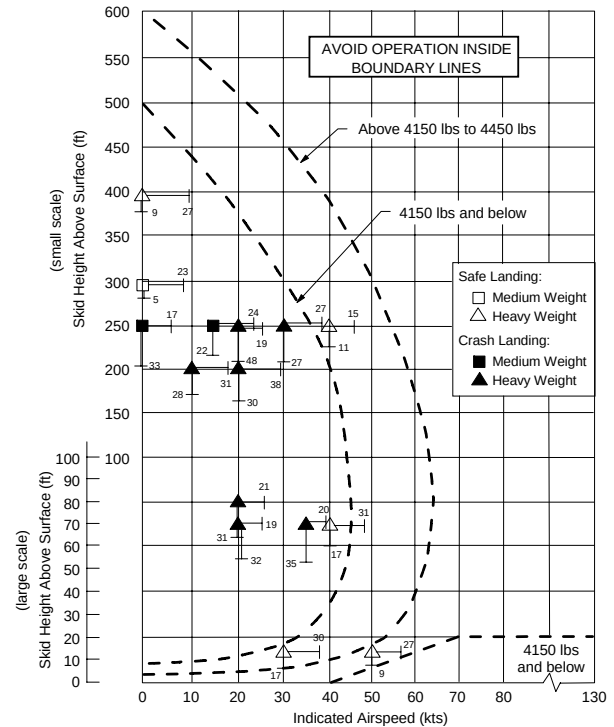


Figure 7. Touchdown ground-speed and sink-rate (medium and heavy weight conditions)

The touchdown sink-rates and forward speeds for all the automated autorotation entry conditions shown in Fig. 5 are presented in Fig. 6 (light-weight condition) and Fig. 7 (medium and heavy weight conditions). Figures 6 and 7 indicate that, in most situations, touchdown conditions were well within the limitations of the rotorcraft (compare touchdown conditions for the crash-landings with the safe landings). Almost all the landings were accomplished with some forward velocity. This is especially true in the heavy and medium weight conditions. This is primarily due to the use of the flare law for the landing. Examination of the optimal solutions for these evaluations indicated that if the helicopter was landed by the optimal algorithm (assuming the technical difficulties were resolved), the forward velocities at touchdown would have been reduced.

Selected time histories for the automated autorotations are presented in Figs. 8, 9, 10, and 11 for the light weight condition and Fig. 12 for the heavy-weight condition. In all these examples, the engine is failed at $t = 0$ and the displayed time history is ended when touchdown is registered by the simulation computer. Figures 8 and 9 demonstrate the extreme nature of the maneuver that is required when autorotating from a hover at 200 ft and 400 ft altitude (above ground level). Figures 8 and 9 demonstrate that it is possible to autorotate safely from well within the avoid region of the H-V curve, if the control inputs are well-timed and of appropriate magnitude. At the lower entry altitude (Fig. 8), immediate nose down pitch attitude of approximately 30 degrees is commanded whereas collective is lowered to zero over a period of roughly 3 sec following engine failure. A pitch pull-up is commenced at an altitude of approximately 100 ft continuing into a landing flare using pitch attitude and collective input at approximately 50 ft altitude. The sharp discontinuity in the longitudinal cyclic at approximately 50 ft altitude marks the transition from the optimal algorithm to the flare law. No attempt was made to refine the algorithm to smoothly transition between these modes, hence the sharp discontinuity. Rotor speed is maintained above 80% throughout most of the maneuver with rotor speed reducing to 60% at touchdown as rotor speed is sacrificed to reduce the touchdown sink rate.

The longer maneuver time allowed by the higher entry altitude is evident in Figure 8. The collective is lowered immediately but there is no attempt to push the nose over and gain airspeed until the rotor-speed approaches

its lower constraint of 75%. To maintain rotor speed above the constraint of 75%, the optimal guidance trades altitude for airspeed and for maintaining rotor speed. A maximum nose-down pitch attitude of 40 degrees is observed in Fig. 9. A run-on landing is achieved at a forward speed of approximately 20 kts (touchdown sink rate of almost zero).

Figure 10 demonstrates the effectiveness of the algorithm for an entry condition in the high-speed "avoid" region of the H-V curve. Due to the low altitude, the flare law almost immediately overrides the optimal algorithm. The helicopter pitches up and trades airspeed for rotor-speed and altitude, placing it in a suitable energy state for a safe flare and touchdown at a forward speed of less than 10 kts. Figure 11 demonstrates an autorotation from an entry condition that is outside the manufacturer's recommended avoid region of the H-V curve for the light-weight condition. In response to the optimal guidance commands, the helicopter initially pitches nose-up to reduce airspeed followed by nose-down pitch to gain airspeed and maintain rotor speed above the constraint limit of 75%. Touchdown is achieved at a sink rate of 3 ft/sec and a forward speed of 40 kts.

The capability of the automatic guidance to safely autorotate for the heavy-weight condition is demonstrated in Fig. 12. The engine is failed when the helicopter is at a hover at an altitude of 400 ft above ground. When contrasted with an autorotation from a similar entry condition for the light-weight condition (Fig. 9), the helicopter sinks more rapidly resulting in a shorter flight time. The optimal guidance commands an almost immediate push-over to gain airspeed (contrast with almost no pitch input for several seconds in Fig. 9) and maintain rotor-speed with a very rapid pull-up to about 35 degrees to arrest sink rate at low altitude. The pull-up results in the rapid increase in rotor-speed to approximately 100% which is traded-off for sink-rate reduction using collective. Touchdown is achieved at a sink rate close to zero and a forward speed of 27 kts. As would be expected the heavier weight conditions proved to be much more challenging in terms of setting the optimal algorithm constraints appropriately and adjusting the flare law due to the rapid sink rate that is established when power is failed leaving very little room for computational or timing errors.

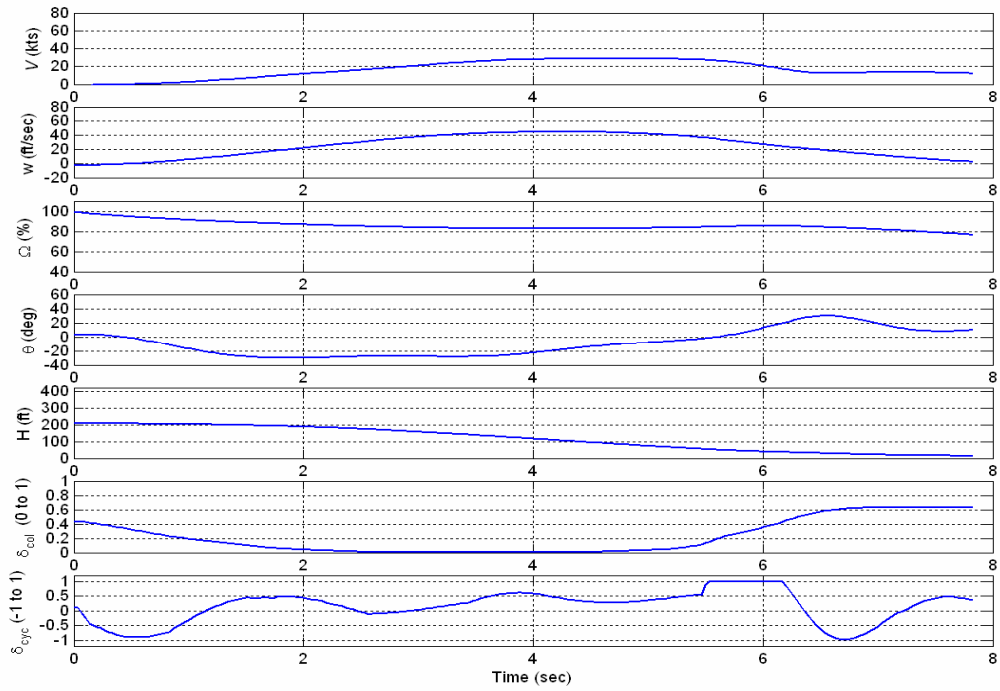


Figure 8. Automatic autorotation from 200ft/0kts; light weight condition (2900 lbs)

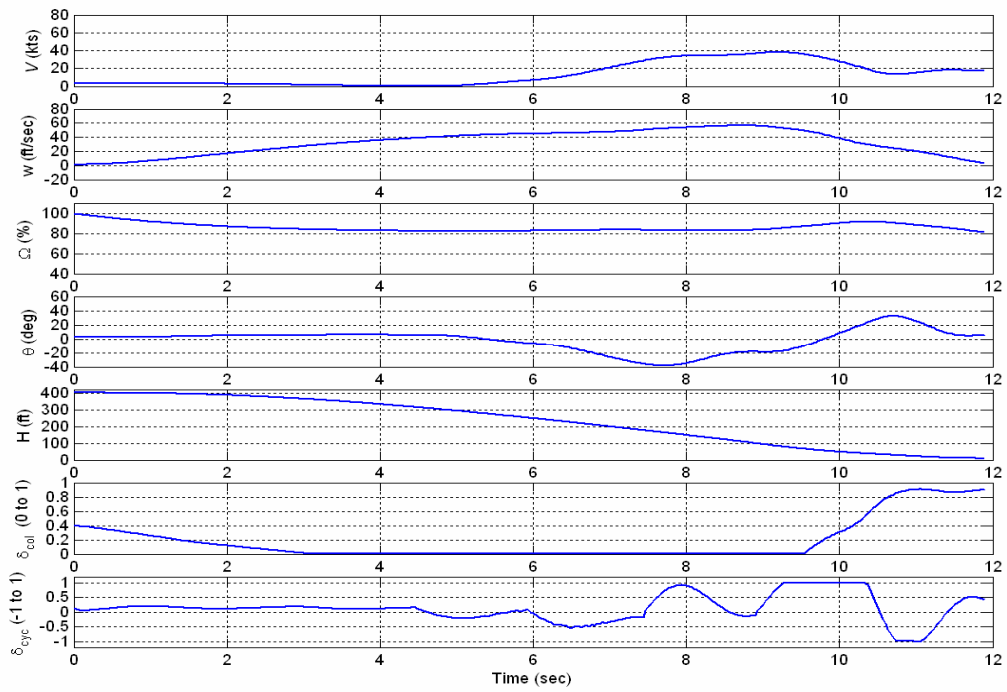


Figure 9. Automatic autorotation from 400ft/0kts; light weight condition (3100 lbs)

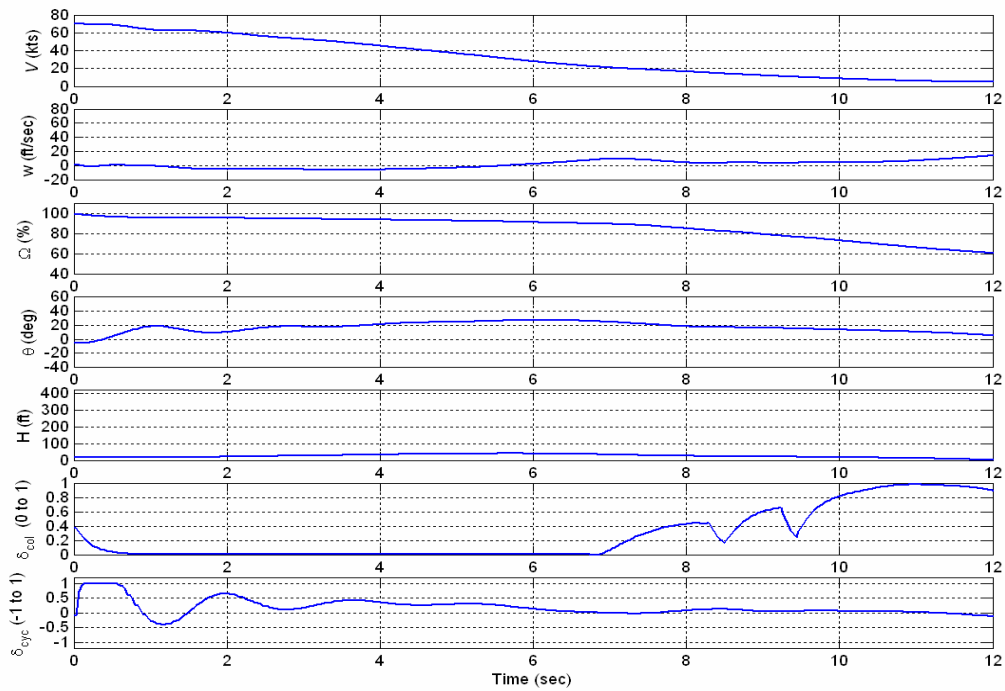


Figure 10. Automatic autorotation from 20ft/70kts; light weight condition (3085 lbs)

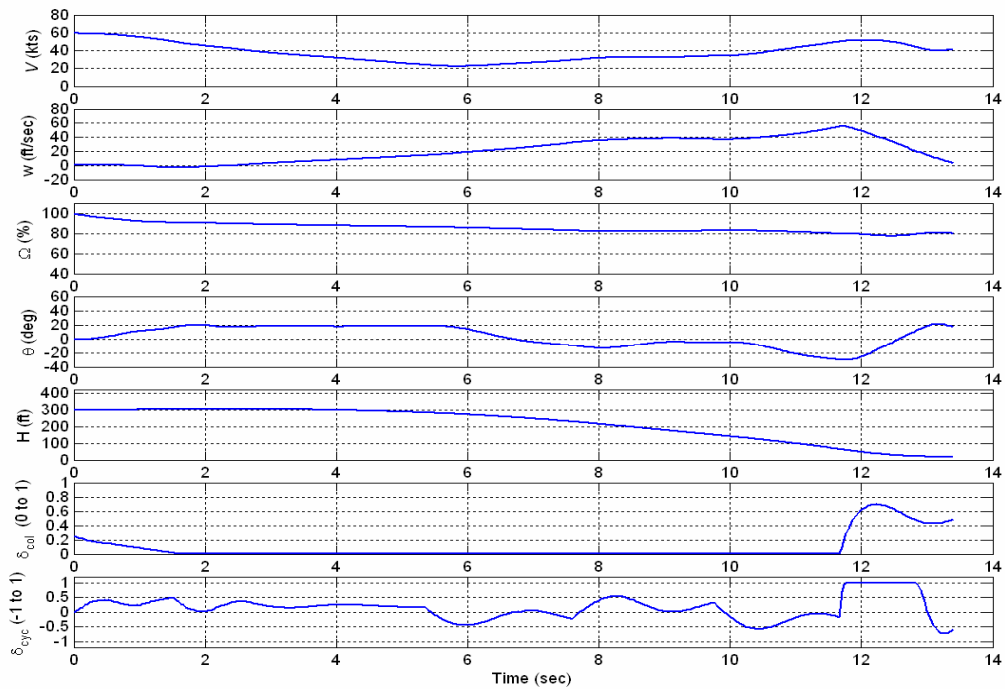


Figure 11. Automatic autorotation from 300ft/60kts; light weight condition (3085 lbs)

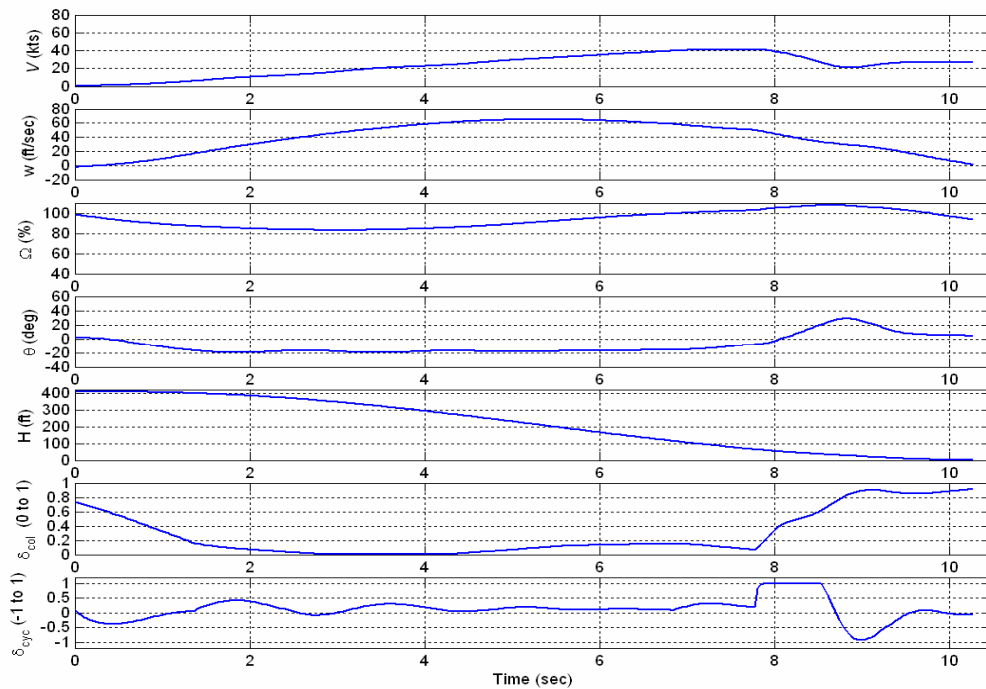


Figure 12. Automatic autorotation from 400ft/0kts; heavy weight condition (4440 lbs)

Conclusions

It has been demonstrated that an appropriately formulated optimization algorithm may be used to provide autorotation guidance in real-time to a rotorcraft. This “automated autorotation” capability will be beneficial on unmanned rotorcraft where redundancy for failure management is not necessarily a primary design requirement. The demonstration was conducted on a high-fidelity commercial flight training device (FTD). When appropriately formulated, the optimal guidance demonstrated a repeatable capability to safely autorotate the helicopter from a variety of entry conditions and a range of weights, even when these entry conditions were well within the avoid region of the height-velocity diagram. Further refinement is necessary to the optimal guidance algorithm to enable it to be robust over a greater flight envelop and weight conditions. A key refinement will be the capability to continuously update the optimal solution throughout the autorotation maneuver. This would compensate for any differences in performance between the rotorcraft and the simple rotorcraft model used in the optimal algorithm and may negate the necessity to include a separate flare law. Further research and development is necessary to investigate these refinements as well as

determine the sensor requirements necessary for implementing this capability in an actual rotorcraft.

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